

Text-RSSR: Text-Conditioned Remote Sensing Super-Resolution with Cross-Modal Learning

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Abstract—This study introduces a Text-Conditioned Remote Sensing Super-Resolution Network (Text-RSSR), which leverages textual and semantic guidance to enhance feature aligning in remote sensing super-resolution tasks. The source code for this work will be accessible at <https://github.com/haoyangofficial/textrssr>. This is the preview version.

Index Terms—Remote sensing image, super-resolution, multimodal learning

I. PROPOSED METHOD

A. Overview

This study introduces a Text-Conditioned Remote Sensing Super-Resolution Network (Text-RSSR) designed to improve remote sensing super-resolution performance through the iterative use of text captions and CLIP embeddings [1]. As shown in Figure 1, the Text-RSSR framework comprises four main steps: Data Preparation, Multimodal Encoding, Text-Guided Iterative Super-Resolution, and Image Decoding.

In the Data Preparation step, a VLM generates a text caption from the high-resolution (HR) image, capturing its detailed content. During Multimodal Encoding, the low-resolution (LR) image is processed by two separate encoders—a CNN Image Encoder and a CLIP Image Encoder—to produce CNN Visual Embeddings and CLIP Visual Embeddings, respectively. Simultaneously, the text caption is processed through a CLIP Text Encoder to generate Text Embeddings. In the CLIP-Guided Iterative Super-Resolution step, the CNN Visual Embeddings are iteratively aligned and refined. In each iteration, all embeddings are input into a Text-Guided Feature Aligning step, followed by processing through a Super-Resolution (SR) Unit. After the final iteration, the refined CNN Visual Embeddings go through one last Text-Guided Feature Aligning. Finally, in the Image Decoding step, the processed CNN Visual Embeddings are decoded by a CNN decoder, upsampled, and further decoded by an additional CNN.

B. Data Preparation

In the Data Preparation step, the HR image captured by the satellite is downsampled to produce an LR image. The

HR image is also used to generate a descriptive text caption. As transferring HR images from satellite to Earth is expensive and demands significant bandwidth, a Vision-Language Model (VLM) is employed onboard the satellite to generate the text caption. To minimize transmission cost and bandwidth usage, only the generated text caption and the LR image are transmitted to Earth.

C. Multimodal Encoding

In this process, the LR image is processed by both a CNN Image Encoder and a CLIP Image Encoder to generate CNN Visual Embeddings and CLIP Visual Embeddings, respectively. At the same time, the text caption is processed through a CLIP Text Encoder to produce Text Embeddings.

As discussed, remote sensing images often contain repetitive patterns and similar scene types, making it difficult for the CLIP Visual Encoder alone to effectively distinguish between them. To address this, a dual image encoder architecture is adopted to combine the semantic understanding capability of the CLIP Visual Encoder with the local feature extraction strength of the CNN. In this architecture, the CNN Visual Embeddings serve as the primary features, while the CLIP Visual Embeddings provide supplementary information to guide feature adjustment and aligning.

The image encoding process is described as follows. Let $\mathbf{L} \in \mathbb{R}^{C \times H \times W}$ represent the LR image received from the satellite. The extraction of CNN Visual Embeddings using the CNN Visual Encoder is defined by:

$$\mathbf{X}_0 = \text{VE}_{\text{CNN}}(\mathbf{L}), \quad (1)$$

where VE_{CNN} denotes the CNN Visual Encoder, and $\mathbf{X}_0$ represents the resulting CNN Visual Embeddings.

To provide CLIP-based semantic guidance, the first token and the remaining tokens from the CLIP Visual Encoder output are separated. A Semantic Guiding (SG) process is then applied to generate a semantic guidance map for each Text-Guided Feature Aligning step in the Text-Guided Iterative Super-Resolution process [2]. The extraction of CLIP Visual Embeddings and the computation of the semantic guidance map for the i -th Text-Guided Feature Aligning step are defined as follows:

$$\mathbf{S}_i = \text{SG}_i(\mathbf{v}_0, \mathbf{v}_{\text{rm}}), \quad (2)$$

where

$$\mathbf{v}_0, \mathbf{v}_{\text{rm}} = \text{VE}_{\text{CLIP}}(\mathbf{L}).$$

Here, $\mathbf{S}_i$ and SG_i denote the CLIP Visual Embeddings and the Semantic Guiding process for the i -th Text-Guided Feature

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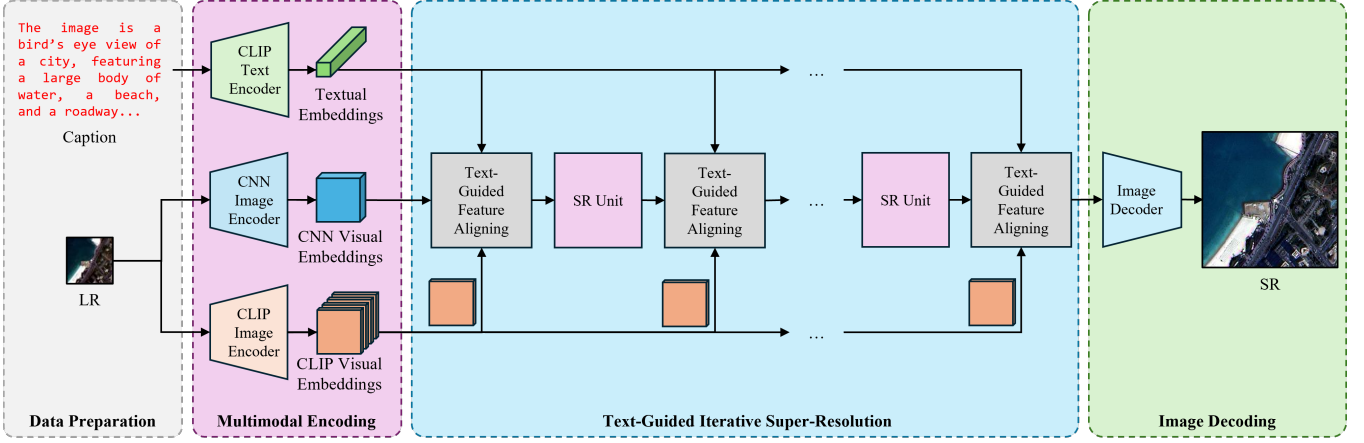


Fig. 1. Overview of the purposed Text-Conditioned Remote Sensing Super-Resolution Network (Text-RSSR).

Aligning step, respectively. VE_{CLIP} denotes the CLIP Visual Encoder, $\mathbf{v}_0$ represents the first token of the resulting CLIP Visual Embeddings, and $\mathbf{v}_{\text{rm}}$ denotes the remaining tokens. Let the number of SR Units be l . Then, the number of Text-Guided Feature Aligning steps is $l + 1$, with the index $i \in [0, l]$.

The SG process is designed to modulate local feature enhancement in the Super-Resolution pipeline using semantic priors from CLIP. For each step $i \in [0, l]$, the SG process computes a semantic guidance map $\mathbf{S}_i \in \mathbb{R}^{B \times H \times W}$ for each of the l SR Units, providing explicit spatially-aware guidance aligned with the target semantics.

Let $\mathbf{v}_0 \in \mathbb{R}^{B \times C}$ denote the global semantic embedding (i.e., the [CLS] token) and $\mathbf{v}_{\text{rm}} \in \mathbb{R}^{B \times (H \times W) \times C}$ the local semantic embeddings extracted from the CLIP Visual Encoder. The SG process for the i -th step is defined as:

$$\text{SG}_i(\mathbf{v}_0, \mathbf{v}_{\text{rm}}) = \text{CS}(\Theta(\mathbf{e}_i), \Theta(\mathbf{v}_{\text{rm}})), \quad (3)$$

where $\text{CS}(\cdot, \cdot)$ denotes the cosine similarity, and $\Theta(\cdot)$ is the ℓ_2 -normalization function applied along the feature dimension. The latent embeddings $\mathbf{e}_i \in \mathbb{R}^{B \times l \times C}$ used for computing the semantic guidance maps are derived through the following stages:

$$\begin{cases} \mathbf{p}_a = \text{MetaNet}(\mathbf{v}_0, \mathbf{p}_0), \\ [\mathbf{p}_a^*; \mathbf{p}_b^*] = \text{MHSA}([\mathbf{p}_a; \mathbf{p}_b]), \\ [\mathbf{e}_i] = \text{MLP}_e(\text{MLP}_a(\mathbf{p}_a^*) \odot \sigma(\text{MLP}_b(\mathbf{p}_b^*))). \end{cases} \quad (4)$$

Here, $\mathbf{p}_0 \in \mathbb{R}^{1 \times C}$ is a learnable global prior descriptor, and $\mathbf{p}_b \in \mathbb{R}^{(l+1) \times C}$ are learnable local unit descriptors, shared across all steps. MetaNet is an architecture to extract semantic correspondence between $\mathbf{v}_0$ and $\mathbf{p}_0$ [3]. The Multi-Head Self-Attention (MHSA) block refines the concatenated descriptors $[\mathbf{p}_a; \mathbf{p}_b]$, enabling context exchange between global and unit tokens [4]. The gating mechanism modulates the unit embeddings via element-wise product between the transformed global vector and a gated local vector. The resulting fused descriptors $\mathbf{e}_i$ are compared with the normalized local features $\mathbf{v}_{\text{rm}}$ via cosine similarity, and $i \in [0, l]$.

The output of the semantic guiding process is a semantic guidance map $\mathbf{S}_i \in \mathbb{R}^{B \times H \times W}$, where $H \times W$ corresponds to the resolution of the LR image. This map serves as a

soft spatial prior for the i -th Text-Guided Feature Aligning step, guiding the model to enhance semantically meaningful regions in accordance with the text prompt. This design allows each SR Unit to adaptively incorporate global semantic context and local alignment cues, thereby improving spatial correspondence and preserving high-frequency details during the Super-Resolution reconstruction process.

The Text Embeddings are generated using the CLIP Text Encoder as follows:

$$\mathbf{T} = \text{TE}(\mathbf{t}), \quad (5)$$

where $\mathbf{t}$ denotes the caption of the HR image generated by the VLM, and $\mathbf{T}$ represents the resulting Text Embeddings.

D. Text-Guided Iterative Super-Resolution

In this step, the CNN Visual Embeddings are iteratively aligned and refined. To maintain the supervisory influence of the CLIP Visual Embeddings and Textual Embeddings throughout the entire Super-Resolution process, each iteration begins by using these embeddings to align the CNN Visual Embeddings. The aligned CNN Visual Embeddings are then processed through an SR Unit. After the final iteration, the refined CNN Visual Embeddings undergo one final Text-Guided Feature Aligning step. Let the CNN Visual Embeddings input for the i -th iteration be $\mathbf{X}_i$, where $i \in [0, l]$. Then, the i -th iteration is defined as:

$$\mathbf{X}_{i+1} = \text{SR}(\text{TGFA}(\mathbf{X}_i)), \quad (6)$$

where TGFA denotes the Text-Guided Feature Aligning step, and SR denotes the Super-Resolution Unit.

Text-Guided Feature Aligning is designed to align and refine the CNN Visual Embeddings using both the CLIP Visual Embeddings and the CLIP Textual Embeddings. In this process, textual information provides channel-wise attention, enabling the model to identify and emphasize the most relevant feature channels for the Super-Resolution task. Following this channel-wise refinement, the semantic guidance maps are applied to perform pixel-wise refinement, offering direct high-frequency guidance to enhance spatial detail in the reconstructed image.

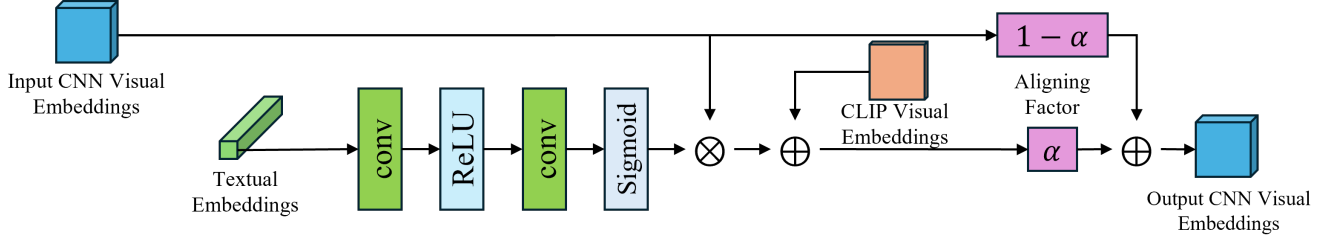


Fig. 2. Overview of the Text-Guided Feature Aligning process.

Let $\mathbf{T} \in \mathbb{R}^d$ be the Textual Embeddings, $\mathbf{S}_i$ the semantic guidance map for the i -th iteration, and d the dimension of CLIP’s hidden state. The Text-Guided Feature Aligning process is defined as:

$$\text{TGFA}(\mathbf{X}_i) = \alpha \cdot \text{Align}(\mathbf{X}_i) + (1 - \alpha) \cdot \mathbf{X}_i, \quad (7)$$

where

$$\text{Align}(\mathbf{X}_i) = \mathbf{X}_i \odot \text{TG}(\mathbf{T}) + \mathbf{S}_i.$$

Here, α is a learnable aligning factor that controls the proportion of alignment, $\text{Align}(\mathbf{X}_i)$ denotes the aligned visual embeddings, and $\odot$ represents element-wise multiplication. The function TG is a lightweight MLP composed of a convolution layer, a ReLU activation, another convolution layer, and a Sigmoid activation, projecting the Textual Embeddings into the channel space.

The entire process of Text-Guided Feature Aligning is illustrated in Figure 2. For the Super-Resolution Unit, we adopt the Residual Token Selective Group (RTSG) from Y. Xiao *et al.*’s Top- k Token Selective Transformer (TTST) [5] architecture.

E. Image Decoding

In the Image Decoding step, the processed CNN Visual Embeddings are used to generate the super-resolved (SR) image, which serves as the final output of the entire network. The CNN Visual Embeddings obtained from the last Text-Guided Feature Aligning are first passed through a CNN, then upsampled, and subsequently processed by another CNN to produce the SR image.

II. EXPERIMENTS

A. Implementation details

This study is implemented using PyTorch [6], and all experiments are conducted on a system equipped with an Intel Core i9-10900X CPU, 128 GiB of RAM, and a single NVIDIA GeForce RTX 4090 GPU. The operating system is Ubuntu 22.04 (Jammy) with CUDA version 12.4. The Text-RSSR framework, along with all baseline models used for comparison, adopts L1 Loss [7] as the loss function and employs the AdamW optimizer [8] with an initial learning rate of 10^{-4} . The learning rate is reduced by half at the midpoint of the training process. The downscale factor used to generate the LR image is set to 4. All models are trained from scratch,

except for the CLIP components. For Text-RSSR, the Vision-Language Model (VLM) used to generate captions is LLaVa-1.5 7B [9]. The CLIP Visual Encoder and Textual Encoder used in the Multimodal Encoding step are pretrained and kept frozen during training. The initial aligning factor α in the Text-Guided Feature Aligning step is set to 0.5. The number of SR Units, l , is set to 6. The RTSG Super-Resolution Unit is initialized following the configuration provided in the original TTST paper [5]. Benchmark evaluations are performed on two public RSSR datasets: Alsat-2B dataset [10], UC Merced Land Use Dataset [11], and Aerial Image Dataset (AID) [12]. Model performance is assessed using Peak Signal-to-Noise Ratio (PSNR) [13] and Structural Similarity Index Measure (SSIM) [14]. Compared to existing methods, Text-RSSR demonstrates strong performance on RSSR tasks.

B. Dataset Description

We conducted experiments on three public RSSR datasets: Alsat-2B dataset [10], UC Merced Land Use Dataset [11], and AID [12]. According to the results, Text-RSSR demonstrates strong performance across all evaluated datasets.

The Alsat-2B dataset, introduced by A. Djerida *et al.*, includes 2,182 training samples and three testing subsets—agriculture, urban, and special scenes—containing 56, 282, and 239 image pairs, respectively. Each high-resolution (HR) image has a fixed resolution of 256×256 , while each low-resolution (LR) image has a resolution of 64×64 . The spatial resolution of the dataset is 2.5 meters. All models are trained for 50 epochs, and the batch size is set to 4.

The UC Merced Land Use Dataset, introduced by Y. Yang *et al.*, consists of 21 categories, each with 100 HR images. Each image has an approximate resolution of 256×256 , and a spatial resolution of 0.3 meters. For consistency in this study, all HR images are reshaped to 256×256 . We randomly select 75 images from each category for training and use the remaining 25 for testing. All models are trained for 200 epochs, and the batch size is set to 4.

The AID comprises 10,000 RGB images, each with a resolution of 600×600 pixels. Introduced by G.-S. Xia *et al.*, it contains 30 scene categories, with several hundred images per class, and a spatial resolution of 0.5 meters. In this study, 80% of the images from each category are randomly selected for training, and the remaining 20% for testing, resulting in 8,000 training images and 2,000 testing images. All HR images are resized to 576×576 , and all LR images are resized to

TABLE I
QUANTITATIVE RESULTS OF THE EXPERIMENT ON ALL DATASETS.

Model	Parameters (M)	FLOPs (G)	Alsatsat-2B		UC Merced Land Use Dataset		AID	
			PSNR (dB)	SSIM	PSNR (dB)	SSIM	PSNR (dB)	SSIM
TTST [5]	18.94	317.68	16.293 4	0.356 1	27.524 3	0.778 0	27.339 3	0.709 3
Text-RSSR	25.15	-	16.363 2	0.370 2	27.554 7	0.778 4	27.400 5	0.711 2

144 × 144. All models are trained for 20 epochs, and the batch size is set to 1.

C. Performance Comparison

We compared the performance of Text-RSSR with one of the state-of-the-art methods from 2024 and 2025: Y. Xiao *et al.*'s TTST [5]. Text-RSSR achieved state-of-the-art performance according to various evaluation metrics, as shown in Table I.

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